Daniel Liu

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AI researcher and engineer working on post-training, evaluation, and agent systems, from research to production.

EXPERIENCE

Founder — AI-native marketing startup · San Francisco, CA 2026

- Founded a startup that paired human strategists with AI-agent workflows for campaign analysis, execution, and reporting. Owned engineering and go-to-market; won multiple paid client engagements.
- Built the multi-agent platform from zero: main-agent / sub-agent orchestration, long-running tool use, persistent memory, a sandboxed execution service, and an in-agent Python analytics sandbox.
- Designed a read-only analysis mode and an approval-gated execution mode that showed a plan before changing any ad account. Connected Meta, Google, LinkedIn, TikTok, and Amazon through OAuth.

Meta — Staff Software Engineer, ML/AI (Software Engineer → Senior → Staff) · Menlo Park, CA 2020–2026

Meta AI for Business — Founding AI Researcher 2024–2026

- Led product direction and core agent-system design for Meta's AI assistant for advertisers, a 100+ person initiative spanning four Meta organizations.
- Built the post-training and evaluation loop end to end: SFT, DPO and GRPO, human and LLM-judge evals, and launch gates.
- Designed production agent infrastructure for multi-step planning, tool use, context management, and multi-agent orchestration.

SMB Ads ML — Tech Lead 2023–2024

- Tech lead for ad delivery, recommendation, and budget-automation systems for small and mid-sized advertisers. Led SMB ads work that drove $100M+ in ad platform revenue.
- Shipped Meta's first ML-powered personalized budget recommendation system for SMB advertisers. Built feed ranking for small-business surfaces end to end.
- Led a workstream of about 10 engineers and 20+ cross-functional partners.

Instagram Ads Ranking & Messenger Kids ML — ML Engineer 2020–2023

- Built computer-vision super-resolution for ad images and video, and reinforcement-learning-based ad-format personalization in ads ranking.
- Built the first ML integrity model for real-time age detection on Messenger Kids, and redesigned the app's connection-ranking models.

Internships — Meta, Speech (2020): PyTorch scripting modules for the open-source Fairseq toolkit. Google, Ads Quality (2019): autoencoder-based ads spam detection.

EDUCATION

University of Illinois Urbana-Champaign — M.S. Computer Engineering (GPA 3.87/4.0), 2020 · B.S. Electrical Engineering, Highest Honors, James Scholar, Dean's List

- Research: controllable outfit generation with Prof. David Forsyth (2018–2020); clinical prediction with the DEPEND group and Mayo Clinic (2017–2019).

PUBLICATIONS & PATENT

- Kedan Li, Chen Liu, David Forsyth. "Coherent and Controllable Outfit Generation." arXiv:1906.07273, 2019.
- "Methods and Systems to Improve Post Opportunities." U.S. Patent Application No. 18/890,540, published as US 2025/0148501 A1. Co-inventor (as Chen Liu).

SKILLS

Python, C++, SQL, TypeScript · PyTorch · Post-training (SFT, DPO, GRPO) · Evaluation design (human and LLM-judge) · LLM agents and multi-agent orchestration · Ranking and recommender systems · Reinforcement learning · Computer vision · Spark, Docker, Kubernetes, FastAPI, AWS